

Question 1 (15 marks)

a) Find $\int \cos^2 x \sin x dx$

2

b) Find $\int \frac{dx}{\sqrt{4x^2 - 36}}$

2

c) Evaluate $\int_0^1 x e^x dx$

3

d) Evaluate $\int_0^3 x^2 \sqrt{x+1} dx$

4

e) Find real numbers a and b such that

(i)
$$\frac{4x^2 + 4x - 4}{(x-1)(x+1)^2} = \frac{a}{x-1} + \frac{b}{x+1} + \frac{2}{(x+1)^2}$$

2

(ii) Hence find $\int \frac{4x^2 + 4x - 4}{(x-1)(x+1)^2} dx$

2

Question 2 (15 marks)

a) Let $z = 2 + 3i$ and $w = 3 - 4i$. Find, in the form $x + iy$,

(i) $\bar{w}$

1

(ii) z^2

1

(iii) $\frac{z}{w}$

1

b) (i) Express $1 + \sqrt{3}i$ in modulus-argument form

2

(ii) Express $(1 + \sqrt{3}i)^8$ in modulus-argument form

2

(iii) Hence express $(1 + \sqrt{3}i)^8$ in the form $x + iy$

1

c) Find, in modulus-argument form, all solutions of $z^3 = 1$

2

d) Sketch the region on the Argand Diagram where the inequalities

3

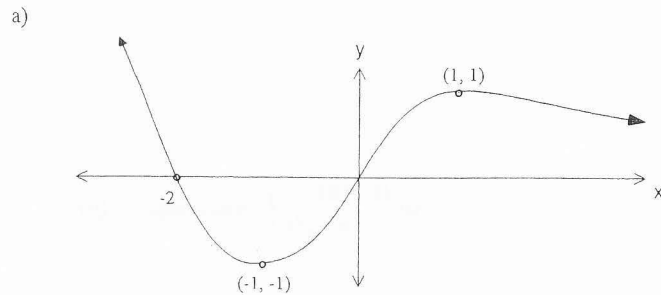
$|z + \bar{z}| \geq 2$ and $|z - 1 - i| < 1$ hold simultaneously

e) Suppose that the complex number z lies on the unit circle, and

2

$0 \leq \arg(z) \leq \frac{\pi}{2}$. Prove that $2 \arg(z-1) = \arg(z) + \pi$

Question 3 (15 marks)



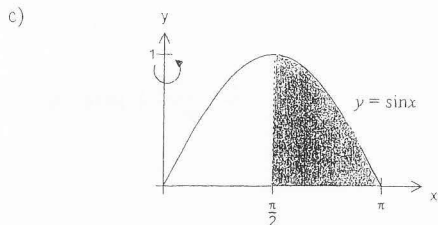
The diagram shows the graph of $y = f(x)$. The x axis is an asymptote. Draw separate one-third page sketches of the following:

- (i) $f(-x)$
- (ii) $f(|x|)$
- (iii) $y = \frac{1}{f(x)}$
- (iv) $y^2 = f(x)$

2
2
2
2

- b) The zeros of $x^3 - 4x^2 + 2x - 1$ are α , β and γ . Find a cubic polynomial with integer coefficients whose zeros are α^2 , β^2 and γ^2 .

3



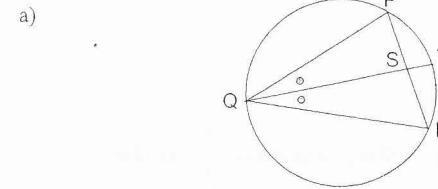
Use the method of cylindrical shells to find the volume of the solid formed when the shaded region bounded by

$$y = 0, y = \sin x, x = \frac{\pi}{2}, x = \pi$$

is rotated about the y -axis

4

Question 4 (15 marks)

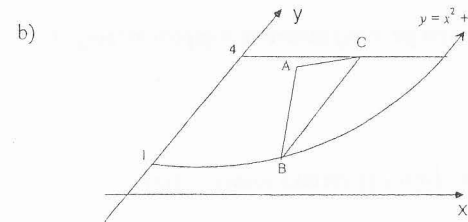


In the diagram, the bisector QT of angle PQR has been extended to intersect the circle PQR at T .

Copy the diagram:

- (i) Prove that the triangles QPS and QTR are similar
- (ii) Show that $QS \cdot QT = QP \cdot QR$
- (iii) Prove that $QS^2 = QP \cdot QR - PS \cdot SR$

2
1
3



3

The base of a solid is the region bounded by the curve $y = x^2 + 1$, the y -axis and the lines $y = 1$ and $y = 4$, as shown in the diagram.

Vertical cross-sections taken through this solid in a direction parallel to the y -axis are equilateral triangles. A typical cross-section, ABC is shown.

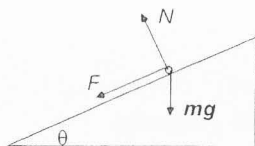
Find the volume of the solid

- c)
- (i) Suppose that a is a double root of the polynomial equation $P(x) = 0$. Show that $P'(a) = 0$.
 - (ii) What feature does the graph of a polynomial have at a root of multiplicity 2?
 - (iii) The polynomial $P(x) = mx^4 - nx^2 + 2$ is divisible by $(x+1)^2$. Find the coefficients m and n .

2
1
3

Question 5 (15 marks)

a)

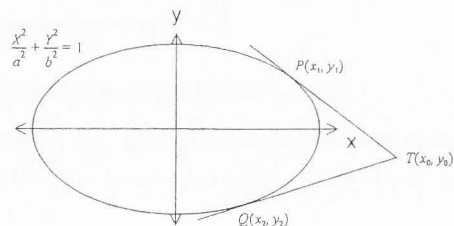


A road contains a bend that is part of a circle of radius r . At the bend, the road is banked at an angle θ to the horizontal. A car travels around the bend at constant speed v . Assume that the car is represented by a point of mass m , and that the forces acting on the car are the gravitational force mg , a sideways friction force F (acting down the road as drawn) and a normal reaction N to the road.

(i) By resolving the horizontal and vertical components of force, find an expression for F 3

(ii) Show that if there is no sideways force $v = \sqrt{gr \tan \theta}$ 2

b)



The points $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

The tangents at P and Q meet at $T(x_0, y_0)$

(i) Show that the equation at P is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ 2

(ii) Hence show that the chord of contact PQ , has equation $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$ 2

(iii) T lies on the directrix of the ellipse. Prove that the chord PQ passes through the focus $S(ae, 0)$ 1

c) (i) Find the equation of the tangent to the curve defined by $x^3 + xy - y^2 = 1$ at the point $(1, 1)$ 3

(ii) Show that the curve in (i) has a stationary point if $9x^4 + 2x^3 + 1 = 0$ 2

Question 6 (15 marks)

a) (i) Prove the identity $\sin(a+b)x + \sin(a-b)x = 2 \sin ax \cos bx$ 1

(ii) Hence find $\int \sin 5x \cos 3x dx$ 2

b) Consider the following statements about a polynomial $P(x)$

(i) If $P(x)$ is odd, then $P'(x)$ is even 1

(ii) If $P'(x)$ is even, then $P(x)$ is odd 1

Indicate whether each of these statements is true or false. Give reasons for your answers.

c) If $z^6 - 1 = 0$

(i) Express all the values of z in modulus argument form 2

(ii) Show that $z^6 - 1 = (z^2 - 1)(z^2 + z + 1)(z^2 - z + 1)$ 1

(iii) Express the roots of $z^4 + z^2 + 1 = 0$ in the form $x + iy$ 3

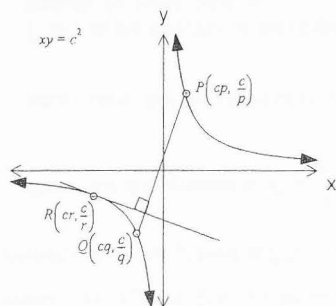
d) (i) Sketch the graph of the function $y = \cos^{-1}\left(\frac{x-1}{2}\right)$ 2

(ii) By adding $y = \sin^{-1} x$ to the graph in (i), solve $\cos^{-1}\left(\frac{x-1}{2}\right) = \sin^{-1} x$ 2

Marks

Question 7 (15 marks)

a)



The points $P(cp, \frac{c}{p})$, $Q(cq, \frac{c}{q})$ and $R(cr, \frac{c}{r})$ lie on the hyperbola $xy = c^2$

The tangent at R is perpendicular to the line joining P and Q .

Show that

(i) The gradient of the tangent at R is $-\frac{1}{r^2}$

(ii) $\angle QRP$ is a right angle

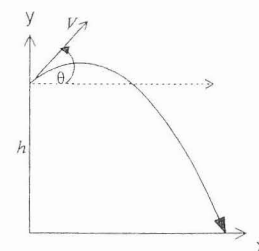
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3

(Question 7 continued on next page)

Question 7 Continued

b)



A projectile is launched from the top of a cliff h metres high with an initial velocity of $V \text{ ms}^{-1}$ at an angle of θ to the horizontal. Given that the horizontal and vertical components of the motion are $\ddot{x} = 0$ and $\ddot{y} = -g$

Show that

(i) $x = Vt \cos \theta$ and $y = Vt \sin \theta - \frac{1}{2}gt^2 + h$

2

(ii) The time of flight, T , is given by

2

$$T = \frac{V \sin \theta + \sqrt{V^2 \sin^2 \theta + 2hg}}{g}$$

(iii) If $h = \frac{V^2 \cos^2 \theta}{2g}$ then the range R of the particle is

3

$$R = \frac{V^2 (\sin 2\theta + 2 \cos \theta)}{2g}$$

c) $S(n) = \log_a x + \log_a x^2 + \log_a x^3 + \dots + \log_a x^n$

(i) Show that $S(n) = \frac{n(n+1) \log_a x}{2}$

2

(ii) Find the value of x if $a = 16$ and $S(100) = 5050$

1

Marks

Question 8 (15 marks)

a) Given that $f(x) = ax^3 + bx^2 + cx + d$

Show that if

(i) $f(x)$ has one stationary point then $b^2 = 3ac$

3

(ii) $f(x)$ has a horizontal point of inflexion then $x = -\frac{c}{b}$

2

b) Given that $I_n = \int_0^\pi x^n \sin x dx$

(i) Show that $I_n = \pi^n - n(n-1)I_{n-2}, n \geq 2$

3

(ii) Evaluate $\int_0^\pi \theta^4 \sin \theta d\theta$

3

c) (i) Using the fact that $A = \frac{1}{2}ab \sin C$ and $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

2

Show that $A = \frac{1}{4}\sqrt{2a^2b^2 + 2a^2c^2 + 2b^2c^2 - a^4 - b^4 - c^4}$

(ii) Hence or otherwise show that the area A of a triangle with sides a, b and c can be found by using the formula.

2

$A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{1}{2}(a+b+c)$

--- End of Exam ---

$$1a) \int \cos^3 x \sin x \, dx = -\frac{1}{3} \cos^3 x + C$$

$$b) \int \frac{dx}{\sqrt{4x^2-36}} = \frac{1}{2} \int \frac{dx}{\sqrt{x^2-9}} = \frac{1}{2} \ln(x + \sqrt{x^2-9}) + C$$

$$c) \int_0^1 x e^x \, dx. \quad \text{let } u=x \quad dv=e^x dx \\ du=dx \quad v=e^x$$

$$\begin{aligned} I &= x e^x - \int e^x dx \\ &= [x e^x - e^x]_0^1 \\ &= e - e - (0 - 1) \\ &= 1 \end{aligned}$$

$$d) \int_0^3 x^2 \sqrt{x+1} \, dx \quad u=x+1 \quad x=0, u=1 \\ du=dx \quad x=3, u=4 \\ (u-1)^2 = x^2$$

$$\begin{aligned} &= \int (u-1)^2 \sqrt{u} \, du \\ &= \int (u^2 - 2u + 1) \sqrt{u} \, du \\ &= \int u^{5/2} - 2u^{3/2} + u^{1/2} \, du \\ &= \left[\frac{2}{7} u^{7/2} - \frac{4}{5} u^{5/2} + \frac{2}{3} u^{3/2} \right]_1^4 \\ &= \left(\frac{2}{7} \cdot 128 - \frac{4}{5} \cdot 32 + \frac{2}{3} \cdot 8 \right) - \left(\frac{2}{7} - \frac{4}{5} + \frac{2}{3} \right) \\ &= \left(\frac{256}{7} - \frac{128}{5} + \frac{16}{3} \right) - \left(\frac{2}{7} - \frac{4}{5} + \frac{2}{3} \right) \\ &= 16 \left(\frac{16}{7} - \frac{8}{5} + \frac{1}{3} \right) - \left(\frac{2}{7} - \frac{4}{5} + \frac{2}{3} \right) \\ &= \frac{1696}{105} \end{aligned}$$

$$e) \frac{4x^2+4x-4}{(x-1)(x+1)^2} = \frac{a}{x-1} + \frac{b}{x+1} + \frac{c}{(x+1)^2}$$

$$4x^2+4x-4 = a(x+1)^2 + b(x-1)(x+1) + c(x-1)$$

$$\text{at } x=1, 4=4a, \quad a=1 \\ x=-1, -4=-4 \quad \checkmark \\ \text{Equate } x^2: 4 = 1+b, \quad \therefore b=3$$

$$\int \frac{4x^2+4x-4}{(x-1)(x+1)^2} dx = \int \left(\frac{1}{x-1} + \frac{3}{x+1} + \frac{2}{(x+1)^2} \right) dx = \ln(x-1) + 3 \ln(x+1) - \frac{2}{x+1} + C$$

$$2a) z = 2+3i, \quad w = 3-4i$$

$$i) \bar{w} = 3+4i$$

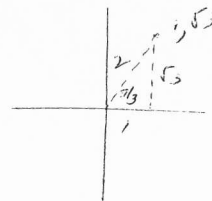
$$ii) z^2 = 4-9+12i = -5+12i$$

$$iii) \frac{z}{w} = \frac{2+3i}{3-4i} \cdot \frac{3+4i}{3+4i} = \frac{6+8i+9i-12}{9+16} = \frac{-6+17i}{25} = -\frac{6}{25} + \frac{17i}{25}$$

$$b) i) 1+\sqrt{3}i = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$ii) (1+\sqrt{3}i)^8 = 2^8 \left(\cos \frac{8\pi}{3} + i \sin \frac{8\pi}{3} \right) = 256 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$iii) = 256 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) = -128 + 128\sqrt{3}i$$



$$i) z^3 = 1$$

$$\text{let } z = \cos \theta + i \sin \theta \\ z^3 = \cos 3\theta + i \sin 3\theta = 1$$

$$\therefore \begin{aligned} 3\theta &= 0, 2\pi, 4\pi \\ \therefore \theta &= 0, \frac{2\pi}{3}, \frac{4\pi}{3} \end{aligned}$$

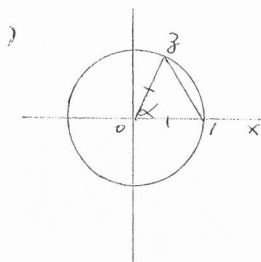
$$\therefore z_1 = \cos 0 + i \sin 0 = 1$$

$$z_2 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$$

$$z_3 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}$$

$$c) |z+\bar{z}| \geq 2, \quad \therefore |2x| \geq 2, \quad |x| \geq 1$$

$$|z-1-i| < 1$$



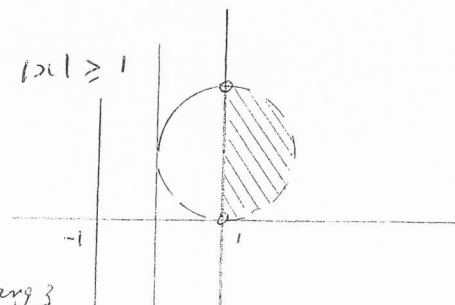
$$\text{let } \angle zOI = \alpha = \arg z$$

$$\therefore \angle zIO = \frac{\pi}{2} - \alpha/2 \quad (\text{isos } \Delta)$$

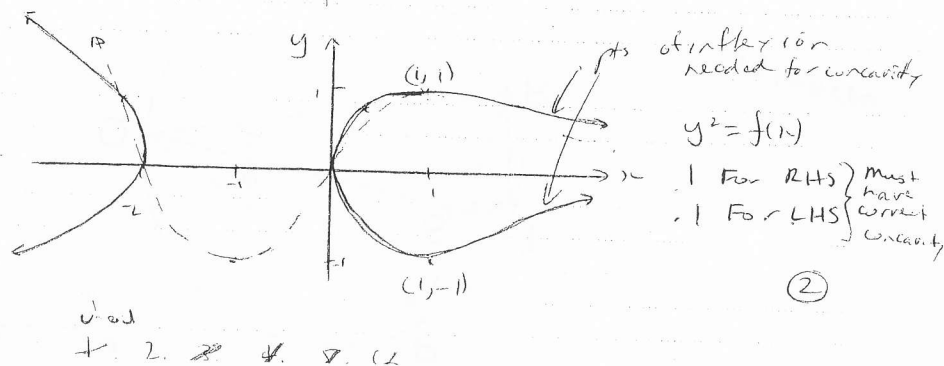
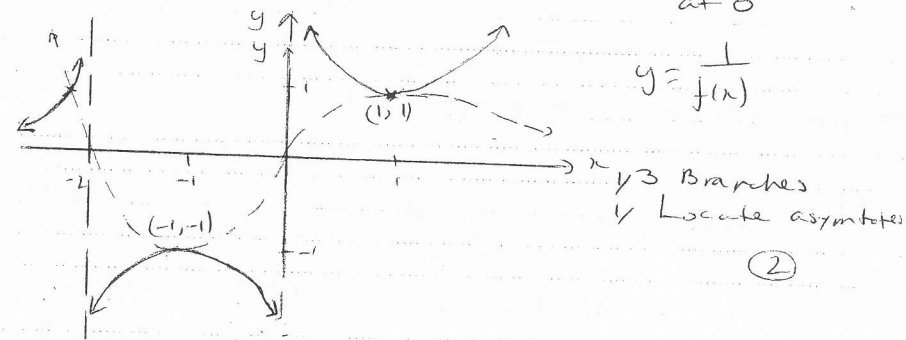
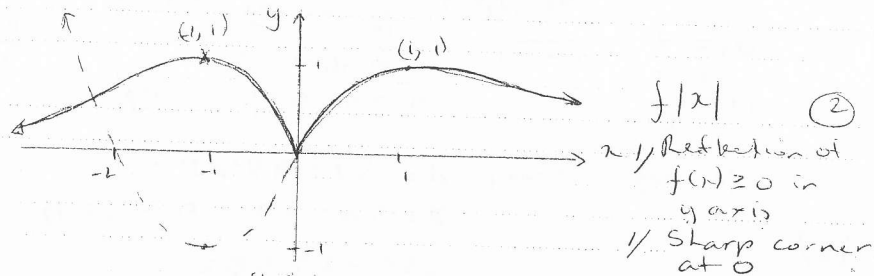
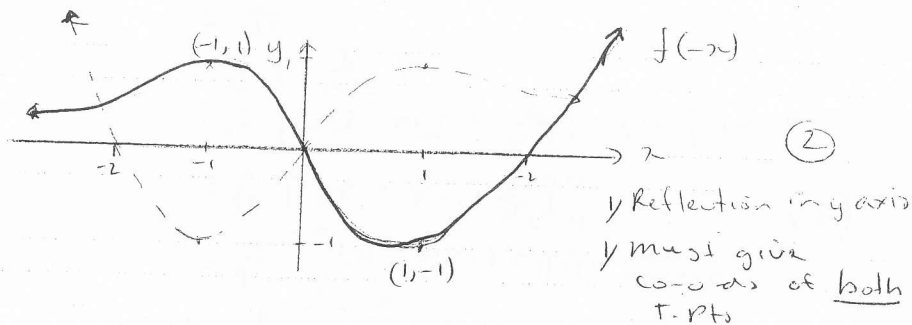
$$\therefore \angle zIx = \frac{\pi}{2} + \alpha/2 \quad (\text{ext. } \angle)$$

$$\text{i.e. } \arg(z-1) = \frac{\pi}{2} + \alpha/2$$

$$\therefore 2 \arg(z-1) = \pi + \alpha = \pi + \arg z$$



Question 3.



(x3)

b) $x^3 - 4x^2 + 2x - 1 = 0$

let $x = \alpha^2, \beta^2, \gamma^2$

ie $\alpha = \sqrt{x}$ sub in above

$$x^3 - 4x^2 + 2x - 1 = 0$$

$$x^2(x+2) = (4x+1)$$

Square both sides

$$x(x^2 + 4x + 4) = 16x^2 + 8x + 1$$

$$x^3 + 4x^2 + 4x = 16x^2 + 8x + 1$$

$$x^3 - 12x^2 - 4x - 1 = 0$$

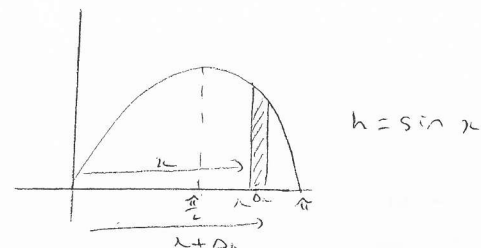
1

2

marks if coefficients are not integers or powers not integers

3

c)



$$V = \pi [(x+\Delta x)^2 - x^2] y$$

$$= \pi [x^2 + 2x\Delta x + (\Delta x)^2 - x^2] y$$

$$= \pi [2x\Delta x] y \quad (\Delta x)^2 \text{ v. small}$$

$$V_{\text{slice}} = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n 2\pi x y \Delta x$$

$$= 2\pi \int_{\frac{\pi}{2}}^{\pi} x \sin x \, dx$$

let $u = x, v = \sin x, u' = 1, v' = -\cos x$

$$V = 2\pi \left[-x \cos x \right]_{\frac{\pi}{2}}^{\pi} + \int_{\frac{\pi}{2}}^{\pi} \cos x \, dx \quad (\text{if 2\pi dropped no penalty})$$

$$= 2\pi \left\{ (\pi - 0) + \left[\sin x \right]_{\frac{\pi}{2}}^{\pi} \right\}$$

$$= 2\pi \{ \pi + (0 - 1) \}$$

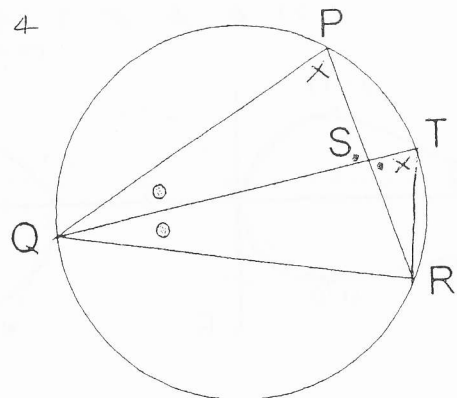
$$= 2\pi [\pi - 1]$$

$$= \underline{2\pi^2 - 2\pi \text{ units}^3}$$

correct evaluation before simplifying (Must be equivalent value)

Question 4

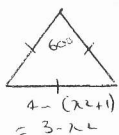
a)



18/1/95/40

- i) Construct TR
In Δ 's QPS, QTR
 $\angle PQS = \angle TQR$ (given)
 $\angle QPS = \angle QTR$ (1 mark) (Angles in same segment)
 $\therefore \Delta QPS \sim \Delta QTR$ (equiangular)
ii) $\frac{QS}{QR} = \frac{QP}{QT}$ (Corresponding sides similar Δ 's) (1 mark)
 $QS \cdot QT = QP \cdot QR$
iii) $PS \cdot SR = QS \cdot ST$ (product of intersecting chords) (1 mark)
 $= QS(QT - QS)$ (1 mark)
 $= QS \cdot QT - QS^2$
 $QS^2 = QS \cdot QT - PS \cdot SR$
 $= QP \cdot QR - PS \cdot SR$ (from ii) (1 mark)

b)



$$V_{\text{slice}} = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} (3 - x^2) \cdot \frac{\sqrt{3}}{2} \cdot x$$

$$V_{\text{solid}} = \int_0^{\sqrt{3}} \frac{\sqrt{3}}{4} (3 - x^2) x dx$$

$$= \frac{\sqrt{3}}{4} \int_0^{\sqrt{3}} (3x - x^3) dx$$

$$= \frac{\sqrt{3}}{4} \left[\frac{3x^2}{2} - \frac{x^4}{4} \right]_0^{\sqrt{3}}$$

$$= \frac{\sqrt{3}}{4} \left(\frac{9}{2} - \frac{9}{4} \right)$$

$$= \frac{\sqrt{3}}{4} \cdot \frac{9}{4}$$

$$= \frac{9\sqrt{3}}{16}$$

Q4

- c) i) Let $p(x) = (x-a)^2 q(x) = 0$
 $p'(x) = 2(x-a)q(x) + (x-a)^2 q'(x)$ (1 mark)
 $= (x-a)[2q(x) + (x-a)q'(x)]$ (1 mark)
 $\therefore p'(a) = 0$
ii) A turning pt (1 mark)
iii) $p(x) = mx^4 - nx^2 + 2$
 $p(1)$ is a root of $p(x)$
 $\therefore m - n + 2 = 0$ (1 mark)
 $p'(x) = 4mx^3 - 2nx$
Now -1 is a root of above
 $\therefore -4m + 2n = 0$ (2 mark)
 $2m - n = 0$ (3)
 $-2m = -4$
 $m = 2$
 $n = 4$ (1 mark)
both correct.

Question Five

a) i) Vertically

$$N \cos \theta = F \sin \theta + mg \quad (1) \quad \underline{1 \text{ mark}}$$

Horizontally

$$N \sin \theta + F \cos \theta = \frac{mv^2}{r} \quad \underline{1 \text{ mark}}$$

From (1) $F \sin \theta = N \cos \theta - mg \times \sin \theta$

" (2) $F \cos \theta = \frac{mv^2}{r} - N \sin \theta \times \cos \theta$

$$F \sin \theta = N \cos \theta \sin \theta - mg \sin \theta \quad (3) \quad (3)$$

$$F \cos \theta = \frac{mv^2}{r} \cos \theta - N \sin \theta \cos \theta \quad (4)$$

$$F = \frac{mv^2}{r} \cos \theta - mg \sin \theta \quad \underline{1 \text{ mark}}$$

only with working

(3) + (4)

ii) Put $F=0$ (1 mark)

$$0 = \frac{mv^2}{r} \cos \theta - mg \sin \theta$$

$$\frac{v^2}{r} \cos \theta = g \sin \theta \quad (2)$$

$$\frac{v^2}{r} = g \tan \theta \quad \underline{1 \text{ mark}}$$

$$v = \sqrt{rg \tan \theta}$$

b) i) $\frac{x_1}{a^2} + \frac{y_1^2}{b^2} = 1$

$$\frac{2x_1}{2a^2} + \frac{2y_1}{b^2} \frac{dy_1}{dx_1} = 0$$

$$\frac{dy_1}{dx_1} = \frac{-b^2 x_1}{a^2 y_1}$$

at P $m = \frac{-b^2 x_1}{a^2 y_1} \quad \underline{1 \text{ mark}}$

Eqn of tangent

$$\frac{y - y_1}{-y_1} = \frac{-b^2 x_1}{a^2 y_1} (x - x_1) \quad \underline{1 \text{ mark}}$$

$$\frac{-y y_1}{b^2} + \frac{y_1^2}{b^2} = \frac{x x_1}{a^2} - \frac{x_1^2}{a^2}$$

$$\frac{x x_1}{a^2} + \frac{y y_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$$

$$\frac{x x_1}{a^2} + \frac{y y_1}{b^2} = 1 \quad (\text{since } (x_1, y_1) \text{ lies on } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1)$$

DSB/11) The tangent at P is $\frac{x_1}{a^2} + \frac{y_1}{b^2} = 1$

and T lies on it $\therefore \frac{x_1 x_0}{a^2} + \frac{y_1 y_0}{b^2} = 1 \quad (1) \quad \underline{1 \text{ mark}}$

In same manner for Q $\frac{x_2 x_0}{a^2} + \frac{y_2 y_0}{b^2} = 1 \quad (2) \quad \underline{1 \text{ mark}}$

from (1) & (2) eqn PQ $\frac{x x_0}{a^2} + \frac{y y_0}{b^2} = 1 \quad (3)$

iii) T $(\frac{a}{2}, y_0)$ sub in (3) $\frac{\frac{a}{2} x_0}{a^2} + \frac{y_0 y_0}{b^2} = 1$

$\frac{x_0}{2a} + \frac{y_0^2}{b^2} = 1 \quad \underline{1 \text{ mark}}$

at S $y=0$ i.e. $\frac{x}{a} = 1 \rightarrow x=a \quad \underline{1 \text{ mark}}$

c) i) $3x^2 + x \frac{dy}{dx} + y - 2y \frac{dy}{dx} = 0 \quad \underline{1 \text{ mark}}$

$$(x-2y) \frac{dy}{dx} = -3x^2 - y$$

$$\frac{dy}{dx} = \frac{-3x^2 - y}{x-2y}$$

at (1,1) $m = \frac{-3-1}{1-2}$

$$= 4 \quad \underline{1 \text{ mark}}$$

eqn of tangent

$$y-1 = 4(x-1) \quad \underline{1 \text{ mark}}$$

$$y = 4x-3$$

ii) for stnry put $\frac{dy}{dx} = 0$

$$\frac{-3x^2 - y}{x-2y} = 0 \quad \underline{1 \text{ mark}}$$

$$y = -3x^2 \text{ subst } x^3 + xy - y^2 = 1$$

$$x^3 + x(-3x^2) - (-3x^2)^2 = 1 \quad \underline{1 \text{ mark}}$$

$$x^3 - 3x^3 - 9x^4 = 1$$

$$9x^4 + 2x^3 + 1 = 0$$

$$x \neq 3, 5, 8$$

$$6a) i) \sin \alpha \cos \theta + \sin \theta \cos \alpha$$

$$+ \sin \alpha \cos \theta - \sin \theta \cos \alpha$$

$$= 2 \sin \alpha \cos \theta$$

$$ii) \frac{1}{2} \int (\sin 8x + \sin 2x) dx$$

$$= -\frac{1}{16} \cos 8x - \frac{1}{4} \cos 2x + C$$

$$b) i) P(x) = a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

$$P'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \dots$$

$$\therefore \text{true}$$

$$ii) P'(x) = a_0 + a_1 x^2 + a_2 x^3 + \dots$$

$$P(x) = a_0 x + \frac{a_1 x^3}{3} + \frac{a_2 x^4}{4} + \dots + C$$

$$\therefore \text{odd on number depending on } C$$

$$\therefore \text{false}$$

$$c) i) \cos 6\theta = 1$$

$$6\theta = 0^\circ, 360^\circ, 720^\circ, 1080^\circ, 1440^\circ, 1800^\circ$$

$$\theta = 0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ$$

$$Z = 120^\circ, \cos 60^\circ, \cos 120^\circ$$

$$\cos 180^\circ, \cos (-120^\circ), \cos (-60^\circ)$$

$$ii) (z^2 - 1)(z^2 + 1) - z^2$$

$$= (z^2 - 1)(z^2 + 2z^2 + 1 - z^2)$$

$$= (z^2 - 1)(z^2 + z^2 + 1)$$

$$= z^2 + z^4 + z^2 - z^2 - z^2 - 1$$

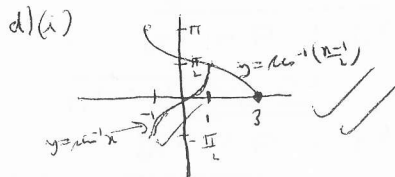
$$= z^4 - 1$$

$$iii) \cos 60^\circ = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\cos 120^\circ = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\cos (-120^\circ) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$\cos (-60^\circ) = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$



$$iv) x = 1$$

$$7a) i) x \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$m_A = -\frac{\frac{y}{x}}{\frac{y}{x}}$$

$$= -\frac{1}{r^2}$$

$$ii) m_{PA} = \frac{\frac{y}{x} - \frac{y}{x}}{cp - cq}$$

$$= \frac{1 - p}{p - q}$$

$$= -\frac{1}{p - q}$$

$$m_{RC} = -\frac{1}{r - q}$$

$$m_{PR} = -\frac{1}{r - p}$$

$$-\frac{1}{r^2} \times -\frac{1}{p - q} = -1$$

$$r^2 p q = 1$$

$$m_{RC} \times m_{PR}$$

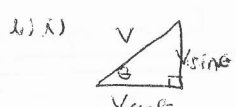
$$= -\frac{1}{r - q} \times -\frac{1}{r - p}$$

$$= \frac{1}{r^2 p q}$$

$$= \frac{1}{-1}$$

$$= -1$$

$$\therefore \triangle RPR \text{ is a right angle}$$



$$x = V \cos \theta$$

$$x = V \sin \theta$$

$$y = -gt + V \sin \theta$$

$$y = -\frac{gt^2}{2} + V \sin \theta t + h$$

$$ii) \frac{g}{L} - V^2 \sin \theta = 0$$

$$t = \frac{V \sin \theta + \sqrt{V^2 \sin^2 \theta - 4 \times \frac{g}{L} \times h}}{2 \times \frac{g}{L}}$$

$$= \frac{V \sin \theta + \sqrt{V^2 \sin^2 \theta + 2gh}}{g}$$

$$iii) R = V t \cos \theta$$

$$= V \cos \theta \times$$

$$\frac{V \sin \theta + \sqrt{V^2 \sin^2 \theta + 2gh}}{g}$$

$$= \frac{V \sin \theta + \sqrt{V^2 \sin^2 \theta + 2gh}}{g} \times V \cos \theta$$

$$= \frac{V \sin \theta + V}{g} \times V \cos \theta$$

$$= \frac{V^2 \sin \theta \cos \theta + V^2 \cos \theta}{g}$$

$$= \frac{2V^2 \sin \theta \cos \theta + 2V^2 \cos \theta}{2g}$$

$$= \frac{V^2 (\sin 2\theta + 2 \cos \theta)}{2g}$$

$$c) i) f(x) = \frac{n}{2} (\log_e x + \log_e x)$$

$$= \frac{n (\log_e x + n \log_e x)}{2}$$

$$= \frac{n \log_e x (1 + n)}{2}$$

$$ii) 50:50 = 100:100 \log_{10} x$$

$$10:100 = 10:100 \log_{10} x$$

$$\therefore \log_{10} x = 1$$

$$x = 10$$

$$8a) i) f'(x) = 0$$

$$3ax^2 + 2bx + c = 0$$

$$x = \frac{-2b \pm \sqrt{(2b)^2 - 4 \times 3a \times c}}{2 \times 3a}$$

$$= \frac{-2b \pm \sqrt{4b^2 - 12ac}}{6a}$$

$$4b^2 - 12ac = 0$$

$$4b^2 = 12ac$$

$$b^2 = 3ac$$

$$ii) f''(x) = 0$$

$$6ax + 2b = 0$$

$$x = -\frac{2b}{6a}$$

$$= -\frac{b}{3a}$$

$$= -\frac{b}{\frac{b^2}{c}}$$

$$= -\frac{c}{b}$$

$$b) i) u = x^n \quad v' = \sin x$$

$$u' = n x^{n-1} \quad v = -\cos x$$

$$I_n = \left[x^n \cos x \right]_0^\pi + n \int_0^\pi x^{n-1} \cos x$$

$$= \pi^n + n \int_0^\pi x^{n-1} \cos x$$

$$u = x^{n-1} \quad v' = \cos x$$

$$u' = (n-1)x^{n-2} \quad v = \sin x$$

$$\int_0^\pi x^{n-1} \cos x = \left[x^{n-1} \sin x \right]_0^\pi - (n-1) \int_0^\pi x^{n-2} \sin x$$

$$= -(n-1) \int_0^\pi x^{n-2} \sin x$$

$$\therefore I_n = \pi^n - n(n-1)I_{n-2}$$

$$ii) \int_0^\pi G^+ \sin G d\theta$$

$$= \pi^+ - 12 I_2$$

$$= \pi^+ - 12 (\pi^2 - 2 I_0)$$

$$= \pi^+ - 12 \pi^2 + 24$$

$$I_0 = \int_0^\pi \sin x dx$$

$$= [-\cos x]_0^\pi$$

$$= 2$$

$$c) i)$$

$$A = \frac{1}{2} ab \sqrt{1 - \left(\frac{a^2 + b^2 - c^2}{2ab} \right)^2}$$

$$= \frac{1}{2} ab \sqrt{\frac{4a^2 b^2 - (a^2 + b^2 - c^2)^2}{4a^2 b^2}}$$

$$= \frac{1}{4} \sqrt{4a^2 b^2 + 2a^2 c^2 + 2b^2 c^2 - a^4 - b^4 - c^4}$$

$$ii) A = \sqrt{\frac{a+b+c}{2} \times \frac{b+c-a}{2} \times \frac{a+c-b}{2} \times \frac{a+b-c}{2}}$$

$$= \sqrt{\frac{a+b+c}{2} \times \frac{a+b-c}{2} \times \frac{c+b-a}{2} \times \frac{c+b-a}{2}}$$

$$= \frac{1}{4} \sqrt{(a^2 + 2ab + b^2 - c^2) \times (c^2 - b^2 + 2ab - a^2)}$$

$$= \frac{1}{4} \sqrt{-(a^2 + b^2 - c^2 + 2ab)(a^2 + b^2 - c^2 - 2ab)}$$

$$= \frac{1}{4} \sqrt{-(a^2 + 2ab + b^2 - c^2)(2a^2 c^2 - 2b^2 c^2 + c^4 - 4a^2 b^2)}$$

$$= \frac{1}{4} \sqrt{4a^2 b^2 + 2a^2 c^2 + 2b^2 c^2 - a^4 - b^4 - c^4}$$